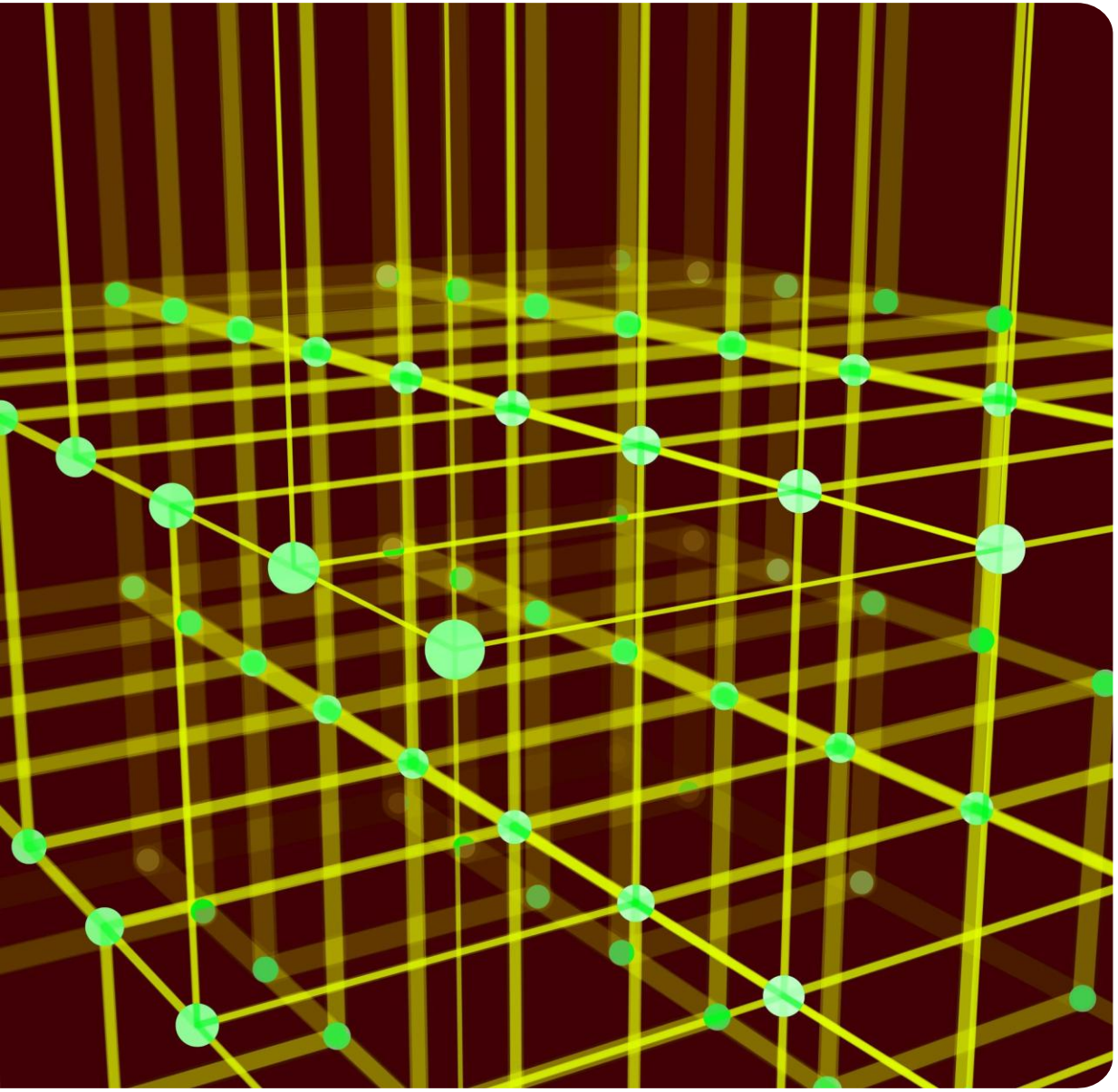
Data Structures and Algorithms I.



University of Szeged

Robert Nagy

2025/26. Autumn Semester

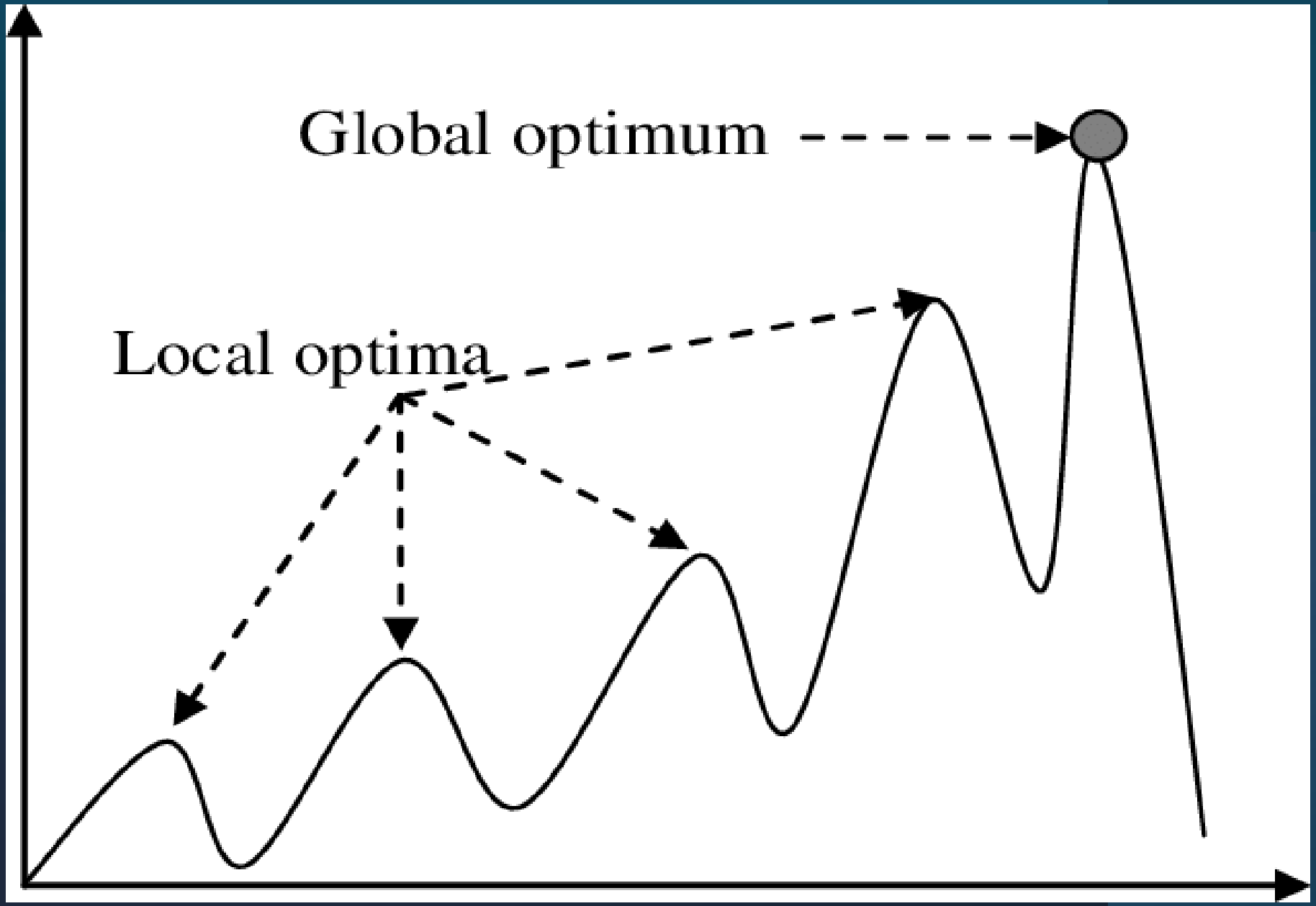


Greedy Algorithms

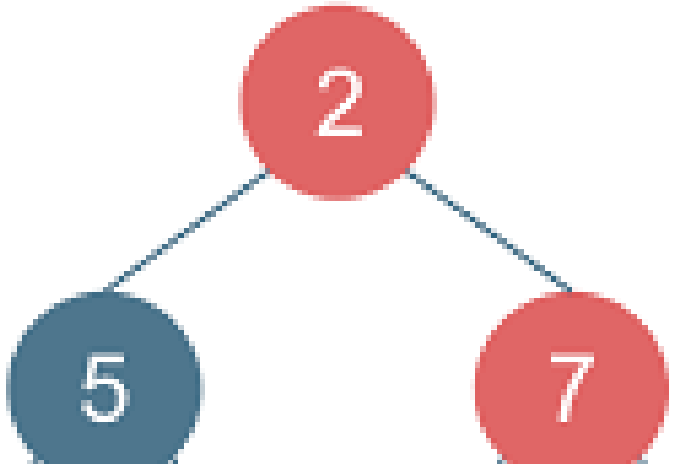
Optimization problems

Global optimum

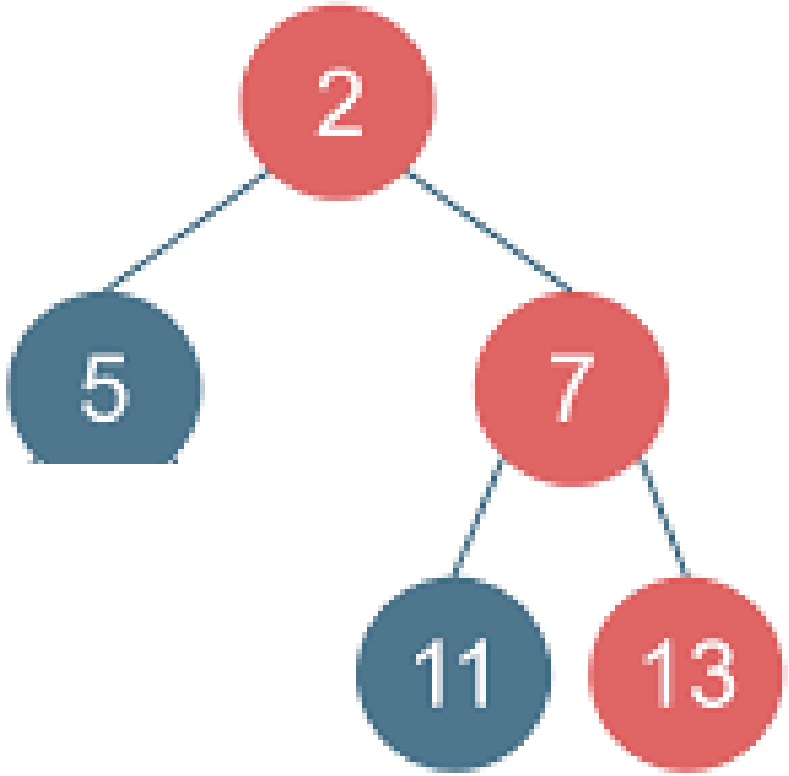
Local optimum


Global optimum
Local optima

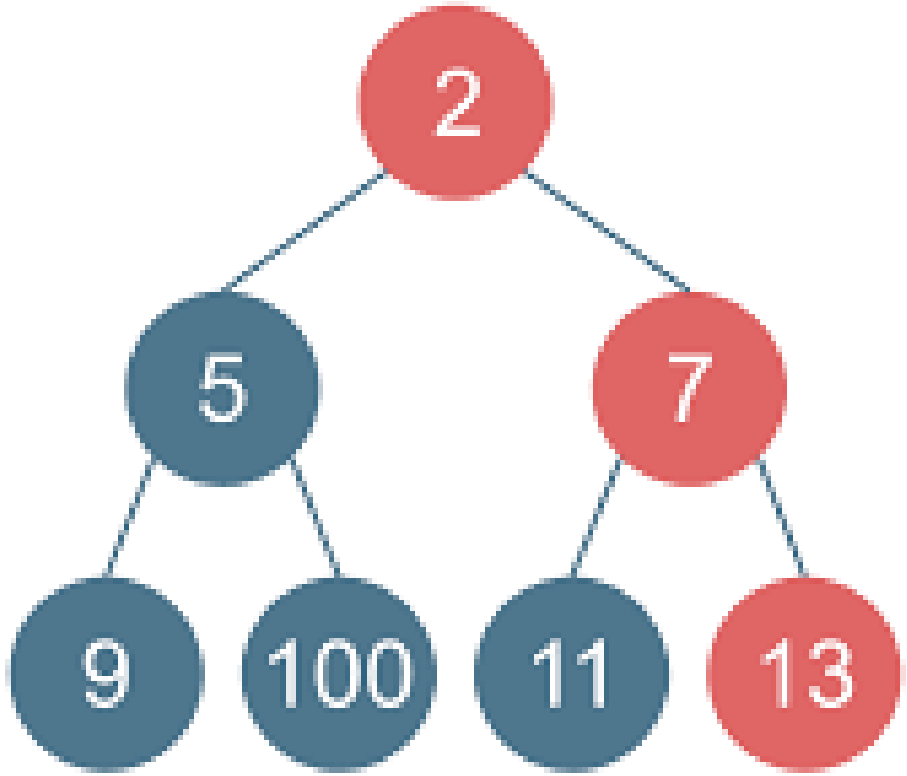
Greedy Solution



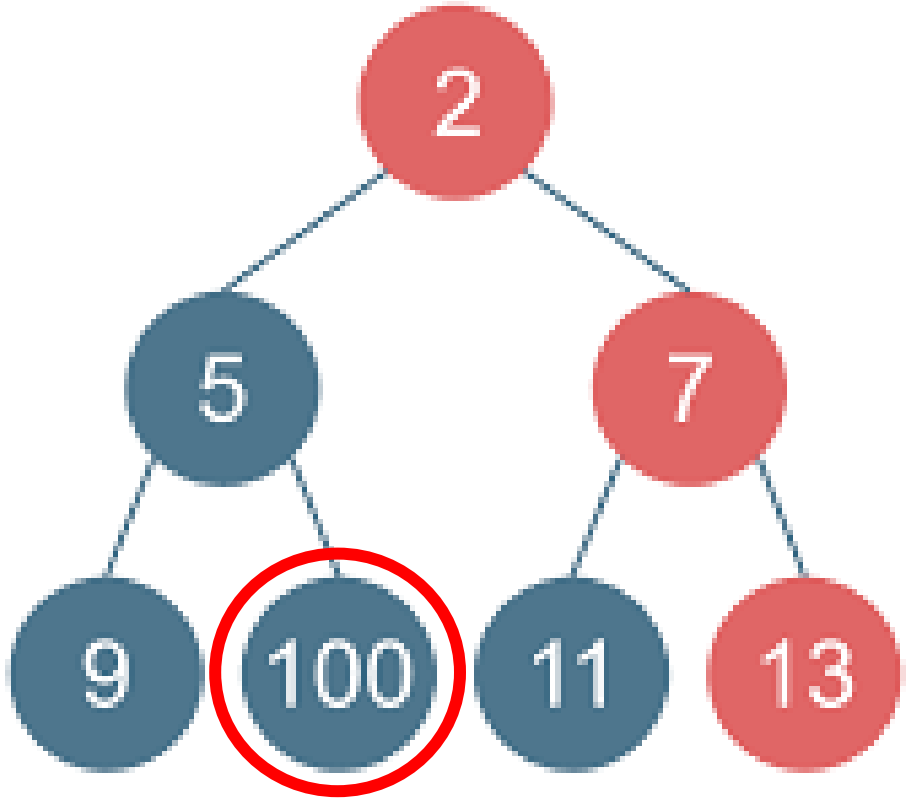
Greedy Solution

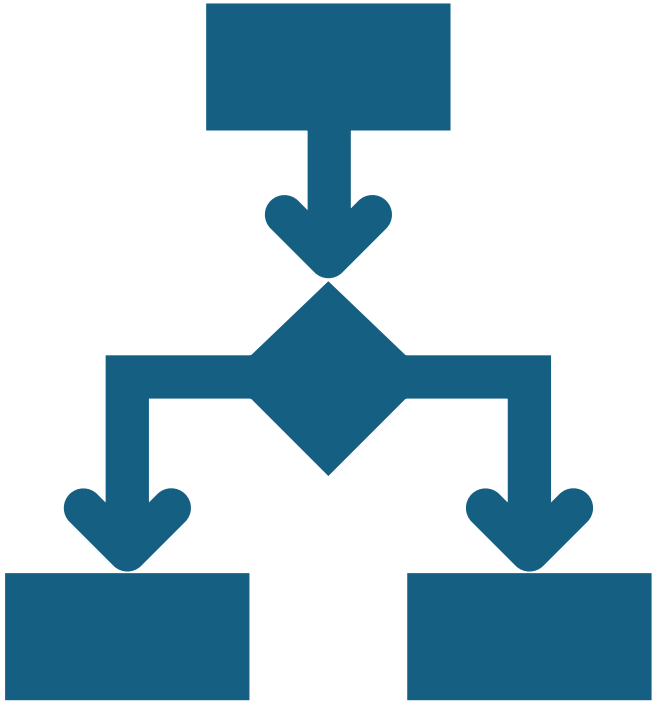


Greedy Solution



Greedy Solution





Top-down approach

We assume the best choice *at the moment* will lead to the best outcome overall

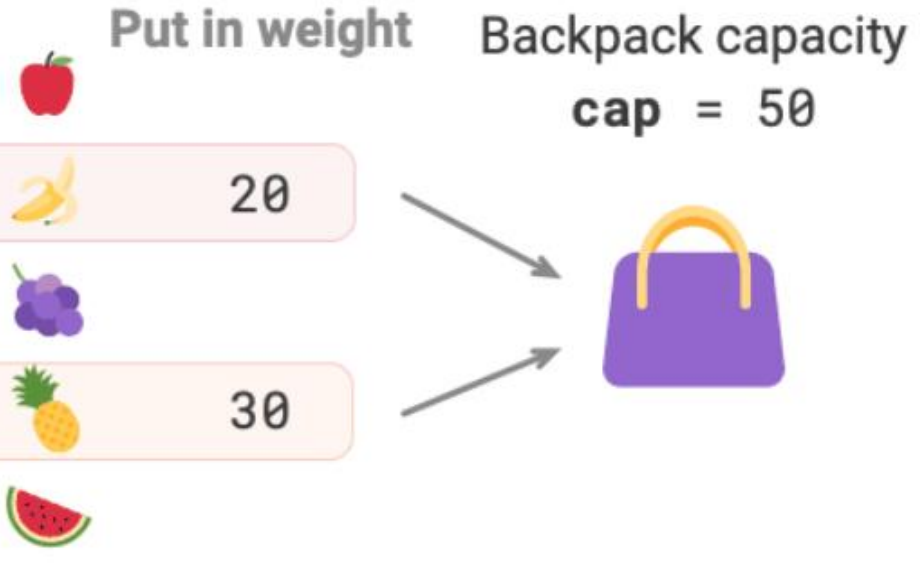
The Fractional Knapsack Problem

https://www.hello-algo.com/en/chapter_greedy/fractional_knapsack_problem/

Index	Weight	Value	
i	wgt[i-1]	val[i-1]	
1	10	50	
2	20	120	
3	30	150	
4	40	210	
5	50	240	

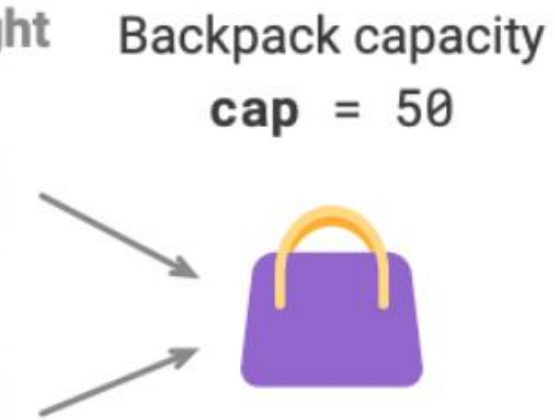
https://www.hello-algo.com/en/chapter_greedy/fractional_knapsack_problem/

Index	Weight	Value
i	wgt[i-1]	val[i-1]
1	10	50
2	20	120
3	30	150
4	40	210
5	50	240



https://www.hello-algo.com/en/chapter_greedy/fractional_knapsack_problem/

Index	Weight	Value
i	wgt[i-1]	val[i-1]
1	10	50
2	20	120
3	30	150
4	40	210
5	50	240



Maximum value: $120 + (30/40) \times 210 = \mathbf{277.5}$

Optimal solution: Put the entire 🍌 and the 🍍 weighing 30 into the backpack, occupying 50 backpack capacity in total

https://www.hello-algo.com/en/chapter_greedy/fractional_knapsack_problem/

Index	Weight	Value	Unit value
i	wgt[i-1]	val[i-1]	$\frac{\text{val[i-1]}}{\text{wgt[i-1]}}$
1	10	50	5
2	20	120	6
3	30	150	5
4	40	210	5.25
5	50	240	4.8



https://www.hello-algo.com/en/chapter_greedy/fractional_knapsack_problem/

Index	Weight	Value	Unit value
i	$wgt[i-1]$	$val[i-1]$	$\frac{val[i-1]}{wgt[i-1]}$
2	20	120	6
4	40	210	5.25
1	10	50	5
3	30	150	5
5	50	240	4.8

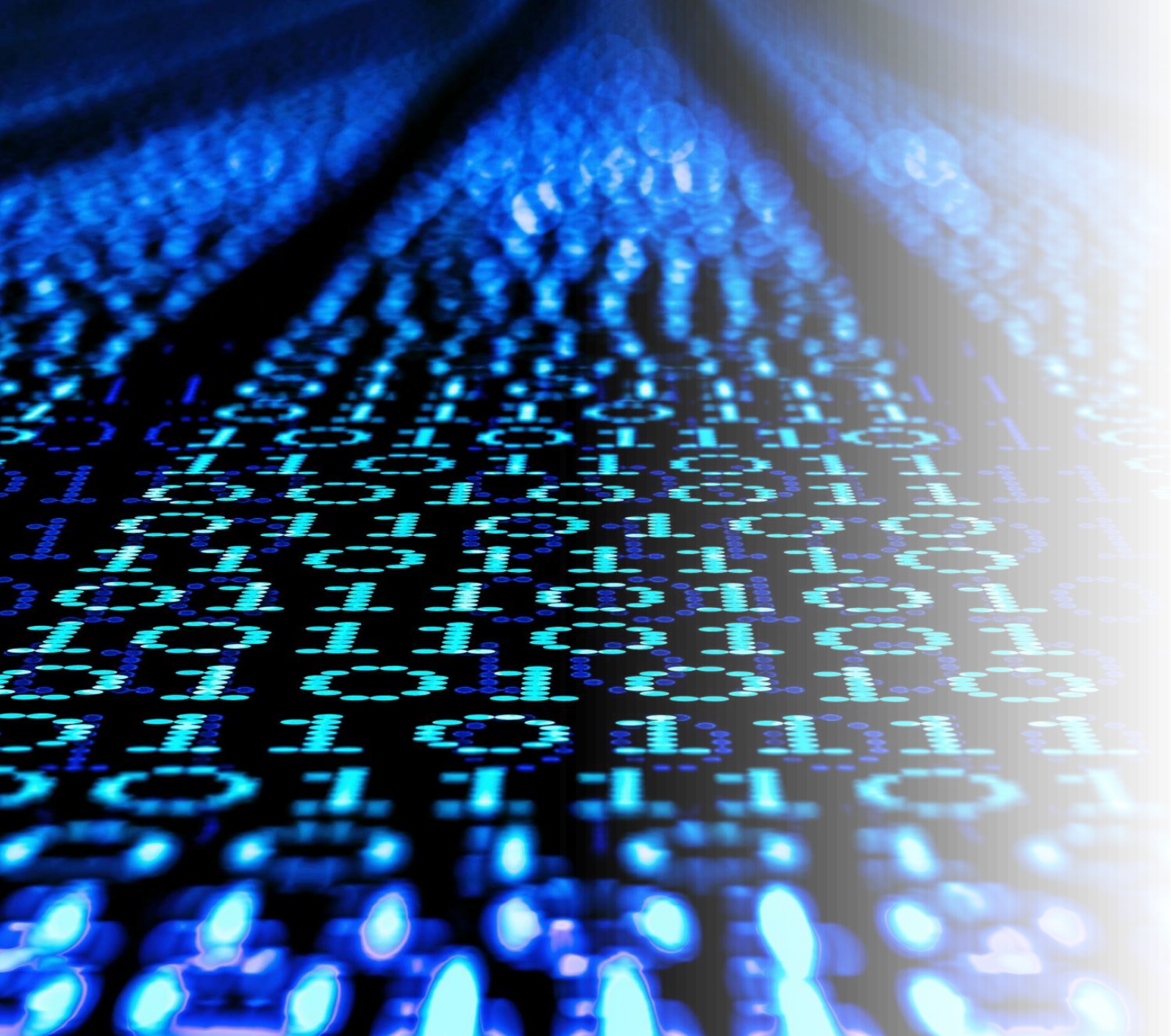
Sort by unit value from high to low



Greedy strategy:
Prioritize choosing items with higher unit value



Huffman codes



Text compression

Suppose the following:

- You have a 100,000-character long message

Suppose the following:

- You have a 100,000-character long message
- Only the following characters are used: a, b, c, d, e, f

Character encoding

	a	b	c	d	e	f
Frequency (in thousands)	45	13	12	16	9	5

Character encoding

	a	b	c	d	e	f
Frequency (in thousands)	45	13	12	16	9	5
Fixed-length codeword	000	001	010	011	100	101

Character encoding

	a	b	c	d	e	f
Frequency (in thousands)	45	13	12	16	9	5
Fixed-length codeword	000	001	010	011	100	101
Variable-length codeword	0	101	100	111	1101	1100

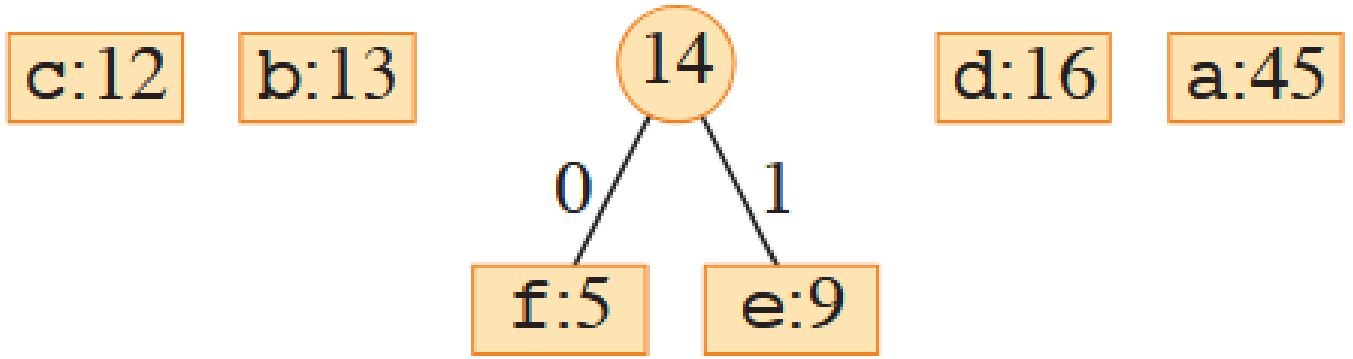
Comparison

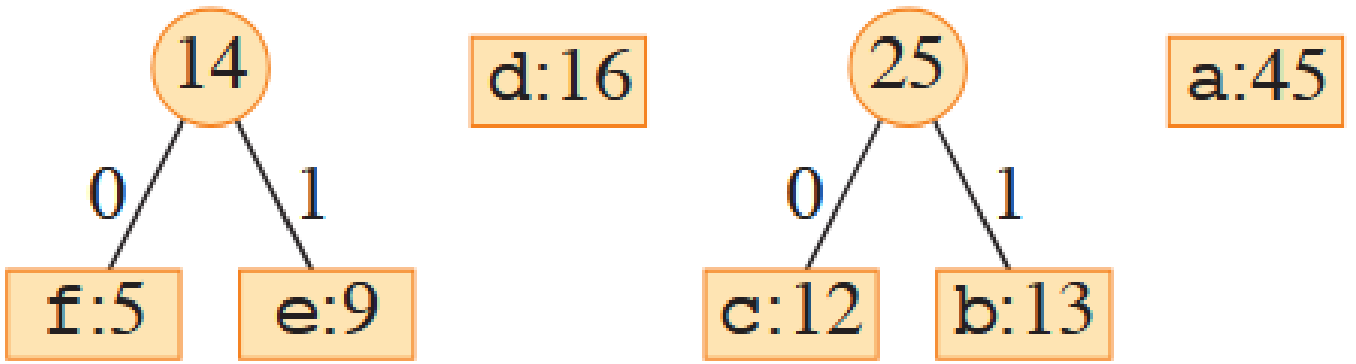
Fixed-length codes – 300,000 bits

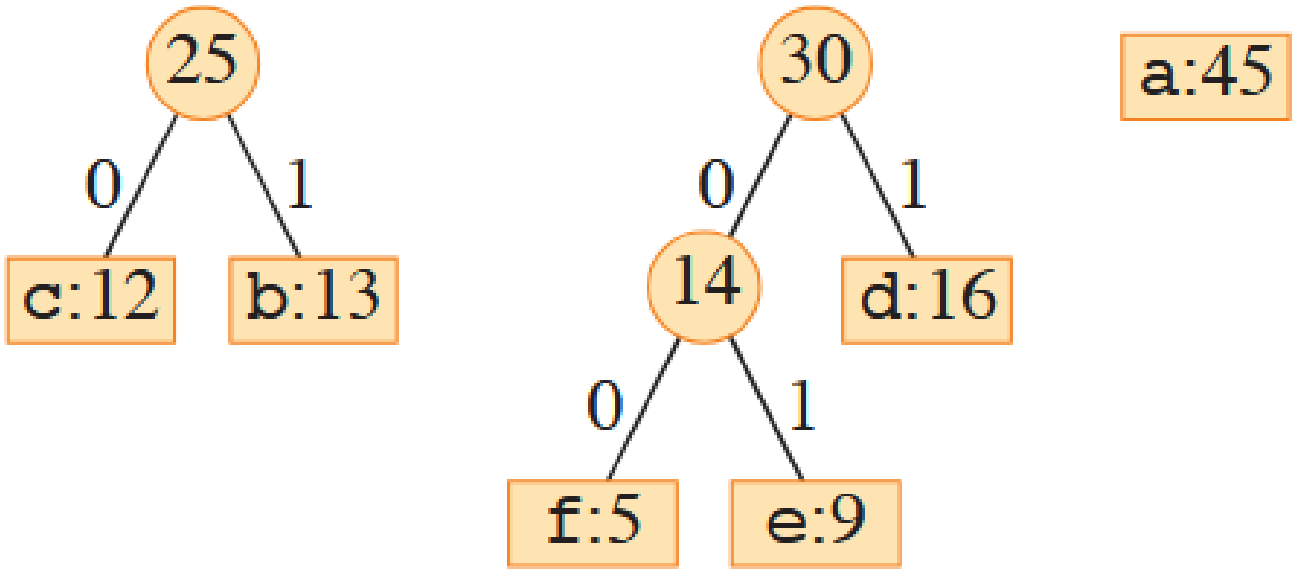
Variable-length codes – 224,000 bits

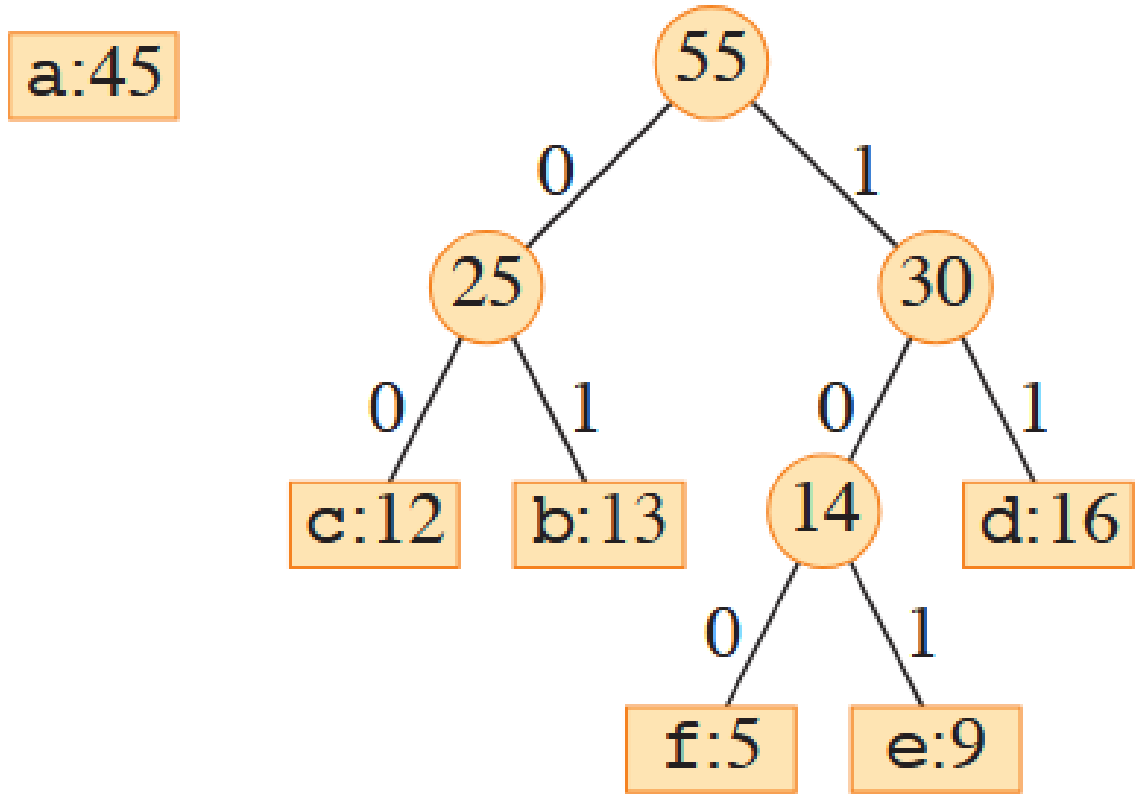
That is an approx. 25% improvement

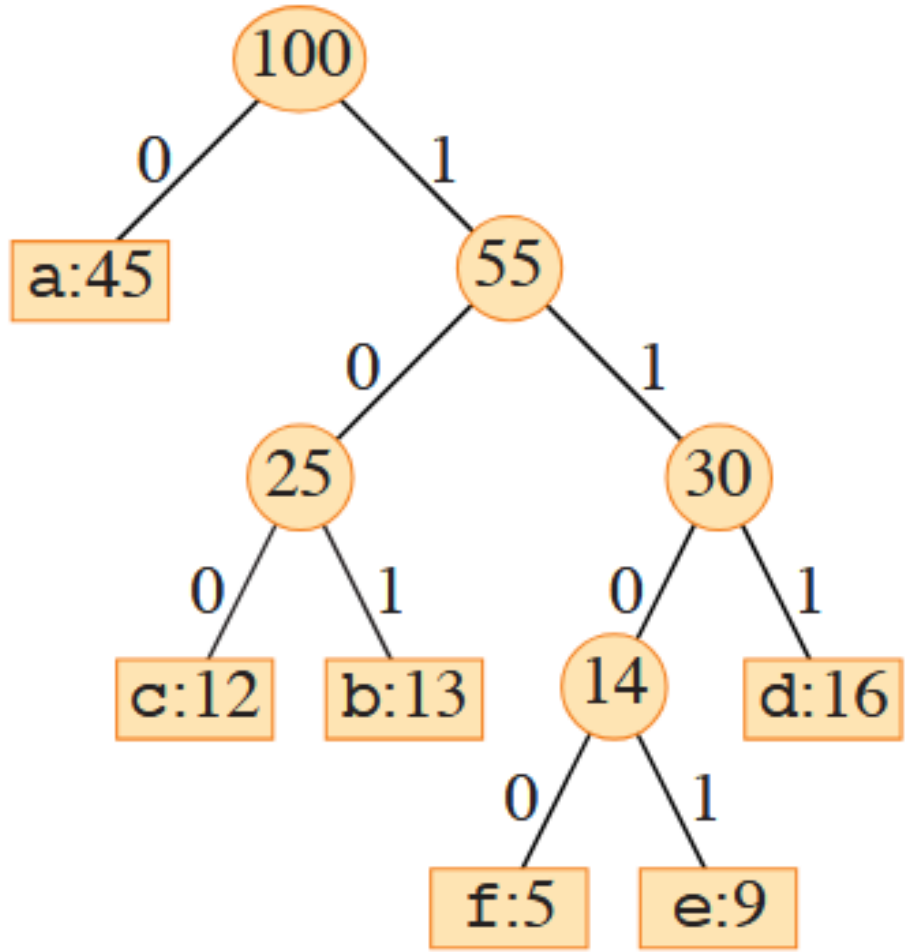

f:5
e:9
c:12
b:13
d:16
a:45


c:12
b:13
14
d:16
a:45
0
1
f:5
e:9


14
0
1
f:5
e:9
d:16
25
0
1
c:12
b:13
a:45


25
0
1
c:12
b:13
30
0
1
14
d:16
0
1
f:5
e:9
a:45


a:45
55
0
1
25
30
0
1
0
1
c:12
b:13
14
d:16
0
1
f:5
e:9


100
0
1
a:45
55
0
1
25
30
0
1
0
1
c:12
b:13
14
d:16
0
1
f:5
e:9

DP *VS* Greedy

Dynamic Programming

- Bottom-up approach
- Global Optimum
- More complex

Greedy Algorithms

- Top-down approach
- Local optimum
- Easier, but not always correct